

HONOURS ANALYSIS 1 EXTRA PROBLEMS

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1. SETS AND INDUCTION

Proposition 1. *For any sets A and B , we have*

$$(A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B).$$

Proof. Equality is obvious if A and B are both empty. Thus, we can suppose that one of A and B is non-empty. We have two inclusions to prove:

$$(A \setminus B) \cup (B \setminus A) \subseteq (A \cup B) \setminus (A \cap B), \quad (A \cup B) \setminus (A \cap B) \subseteq (A \setminus B) \cup (B \setminus A).$$

- (1) We show the inclusion on the left. Let $x \in (A \setminus B) \cup (B \setminus A)$ so that it belongs to $(A \setminus B)$ or $(B \setminus A)$. Without loss of generality, we can assume that $x \in A \setminus B$.¹ Then, $x \in A$ but $x \notin B$. This means that

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B.$$

Hence, $x \in (A \cup B) \setminus (A \cap B)$.

- (2) Conversely, let $x \in (A \cup B) \setminus (A \cap B)$. This simply means that

$$x \in A \cup B \quad \text{and} \quad x \notin A \cap B.$$

Without loss of generality, we can assume that $x \in A$. Since $x \notin A \cap B$, we cannot have $x \in B$. Therefore, $x \notin B$. This implies that $x \in A \setminus B$. More generally,

$$x \in (A \setminus B) \subseteq (A \setminus B) \cup (B \setminus A).$$

This completes the proof. □

Proposition 2. *Prove that*

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Proof. For $n \in \mathbb{N}$, the proposition/statement depending on n we wish to prove is the equality

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n.$$

¹Otherwise, $x \in B \setminus A$. In this case, we just relabel A as B and B as A !

Since we wish to prove the above for all $n \geq 1$, our base case should be $n = 1$.

Base case. We verify directly that the statement is true for $n = 1$:

$$3 = 8 \cdot 1 - 5 = 4 \cdot 1^2 - 1.$$

Inductive step. Our induction hypothesis will be that the statement is true for some $n \geq 1$. Our job is then to show that the claim holds for $n + 1$. The induction hypothesis means that for this n :

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n. \quad (1.1)$$

We now deduce that:

$$3 + 11 + \cdots + (8(n + 1) - 5) = 4(n + 1)^2 - (n + 1).$$

First, our inductive hypothesis in (1.1) gives

$$\begin{aligned} 3 + 11 + \cdots + (8(n + 1) - 5) &= \underbrace{3 + 11 + \cdots + (8n - 5)}_{=4n^2-n} + (8(n + 1) - 5) \\ &\stackrel{(1.1)}{=} 4n^2 - n + 8(n + 1) - 5 \\ &= 4n^2 + 7n + 3. \end{aligned}$$

On the other hand,

$$\begin{aligned} 4(n + 1)^2 - (n + 1) &= 4(n^2 + 2n + 1) - n - 1 \\ &= 4n^2 + 8n + 4 - n - 1 \\ &= 4n^2 + 7n + 3. \end{aligned}$$

Thus,

$$3 + 11 + \cdots + (8(n + 1) - 5) = 4n^2 + 7n + 3 = 4(n + 1)^2 - (n + 1).$$

This completes the inductive step. □

Proposition 3. $2n + 1 \leq 2^n$ for all integers $n \geq 3$.

Proof. This looks like something we should try to prove with induction.

Base case. With $n = 3$, we check directly that

$$2(3) + 1 = 7 < 8 = 2^3.$$

This means that the base case is true.

Inductive step. Suppose that the statement is true for some $n \geq 3$; we must prove that it is also true for $n + 1$. Namely, our induction hypothesis is:

$$2n + 1 \leq 2^n. \quad (1.2)$$

We want to show that $2(n+1) + 1 \leq 2^{n+1}$. First, write

$$2(n+1) + 1 = 2n + 2 + 1 = (2n+1) + 2 \stackrel{(1.2)}{\leq} 2^n + 2. \quad (1.3)$$

In the last step, we used the induction hypothesis (1.2) to get $2n+1 \leq 2^n$. Since $n \geq 3$, it is clear that $2 \leq 2^n$. Therefore, (1.3) implies that

$$2(n+1) + 1 \leq 2^n + 2 \leq 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}.$$

This completes the inductive step and the claim is proven. $\square$

Using this proposition, we prove a far more interesting identity.

Proposition 4. *If $n \geq 5$ is an integer, then $n^2 < 2^n$.*

Proof. Since we are asked to prove an equality involving integers, it is a good idea to try and use mathematical induction. We start with the base case $n = 5$.

Base case. With $n = 5$ we check that

$$5^2 = 25 < 32 = 2^5.$$

Hence, the claim holds for $n = 5$.

Inductive step. Suppose the claim is true for $n \geq 5$, we must show that it is also true for $n+1$. Our inductive hypothesis is therefore the following:

$$n^2 < 2^n. \quad (1.4)$$

Using this, we must show that

$$(n+1)^2 < 2^{n+1}.$$

Using the induction hypothesis (1.4), we calculate

$$(n+1)^2 = n^2 + 2n + 1 \stackrel{(1.4)}{<} 2^n + (2n+1). \quad (1.5)$$

Because $n \geq 5 \geq 3$, we can apply the previous problem to (1.5) above. Doing so gives $(2n+1) \leq 2^n$ whence

$$(n+1)^2 < 2^n + (2n+1) \leq 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}.$$

This completes the inductive step. $\square$

1.1. Power Set.

Proposition 5. *Let X be a set containing exactly n -elements, for any $n \in \mathbb{N}_0$. Then, $|\mathcal{P}(X)| = 2^n$.*

Proof. As this is a statement depending on $n \in \mathbb{N}_0$, it is a good idea to prove the claim by induction.

Base case(s). In the previous examples we verified directly the cases $n = 0$ and $n = 1$. Indeed, the only set containing 0-elements is $\emptyset$.

Inductive Step. We now make the induction hypothesis, suppose that the statement holds for $n \in \mathbb{N}_0$. That is, suppose that $|\mathcal{P}(X)| = 2^n$ for all sets X having n -elements. We want to show that $|\mathcal{P}(Y)| = 2^{n+1}$ for all sets Y having $(n+1)$ -elements. If $n = 0$, then $n + 1 = 1$. By the previous example, we know that the statement is true for $n + 1 = 1$. Therefore, we can assume without loss of generality that $n \geq 1$.

Let now Y be a set with $(n + 1)$ -elements. To count the elements of $\mathcal{P}(Y)$ is to count the subsets of Y . Fix a point $y \in Y$ and define $X := Y \setminus \{y\}$; this set X has n -elements. There are **exactly** two types of subsets of Y :

- (1) subsets of Y that contain y ;
- (2) subsets of Y that do not contain y .

Clearly,

$$\begin{aligned} |\mathcal{P}(Y)| &= \#\text{subsets of } Y \\ &= \#\text{subsets of } Y \text{ that contain } y + \#\text{subsets of } Y \text{ not containing } y. \end{aligned}$$

Let $A \subseteq Y$ be a subset *not* containing y . Then, $A \subseteq X = Y \setminus \{y\}$. By our induction hypothesis, $|\mathcal{P}(X)| = 2^n$. Thus, there are exactly 2^n -subsets of X or, equivalently, 2^n -subsets of Y that don't contain y .

On the other hand, suppose that B is a subset of Y that does contain y . Then, $B \setminus \{y\}$ does not contain y . By the previous part, there are exactly 2^n -distinct possibilities for $B \setminus \{y\}$. But this means that there are precisely 2^n -distinct possibilities for B . Hence, there are 2^n -subsets of Y that contain y . This implies that

$$\begin{aligned} |\mathcal{P}(Y)| &= \#\text{subsets of } Y \text{ that contain } y + \#\text{subsets of } Y \text{ not containing } y \\ &= 2^n + 2^n = 2^{n+1}. \end{aligned}$$

This completes the inductive step. □

2. COUNTABILITY

Problem 1. *Let $\mathcal{F}(\mathbb{N})$ denote the set of all finite subsets of $\mathbb{N}$. Then, $\mathcal{F}$ is countably infinite.*

Proof. There are many ways to prove this, but my favorite is to realize that $\mathcal{F}(\mathbb{N})$ is precisely $\bigcup_{m=1}^{\infty} \mathcal{P}(\mathbb{N}_m)$, where $\mathbb{N}_m = \{1, \dots, m\}$. Thus, as a countable union of finite sets we are done. $\square$

3. SUPREMUM

Problem 2. Show that if a non-empty set $X \subseteq \mathbb{R}$ contains an upper bound, then this upper bound is equal to $\sup X$.

Problem 3. Let X and Y be non-empty subsets of $\mathbb{R}$.

- (1) If both X and Y are bounded from above, then $\sup(X \cup Y)$ exists and is equal to $\max(\sup X, \sup Y)$.
- (2) If both X and Y are bounded from below, then $\inf(X \cup Y)$ exists and is equal to $\min(\inf X, \inf Y)$.

Proof. We only prove the second statement, as the first follows from a similar argument. Since $\inf X \leq x$ and $\inf Y \leq y$ for all $x \in X$ and $y \in Y$, it is clear that $\min(\inf X, \inf Y)$ is a lower bound for the non-empty set $X \cup Y$ (both X and Y are non-empty). Hence, $\inf(X \cup Y)$ exists. Let v be a lower bound for $X \cup Y$, we will show that

$$v \leq \min(\inf X, \inf Y).$$

Because $\min(\inf X, \inf Y)$ is a lower bound for $X \cup Y$, this would mean that $\min(\inf X, \inf Y)$ is the greatest lower bound for $X \cup Y$. Since v is a lower bound for $X \cup Y$, we have $v \leq z$ for all $z \in X \cup Y$. In particular, $v \leq x$ and $v \leq y$ for all $x \in X$ and $y \in Y$. This implies that $v \leq \inf X$ and $v \leq \inf Y$. Therefore, $v \leq \min(\inf X, \inf Y)$. $\square$

3.1. More About Functions. Let X be a non-empty set and $f : X \rightarrow \mathbb{R}$ a function. We say that f is bounded from above if $f(X)$ is bounded from above in $\mathbb{R}$. Similarly, f is said to be bounded from below if $f(X)$ is bounded from below in $\mathbb{R}$. If f is bounded from above, we define

$$\sup_{x \in X} f := \sup f(X) = \sup \{y : y = f(x) \text{ for some } x \in X\}.$$

If f is bounded from below, we have the analogous definition:

$$\inf_{x \in X} f := \inf f(X) = \inf \{y : y = f(x) \text{ for some } x \in X\}.$$

Problem 4. Let X and Y be non-empty sets. Let $f : X \rightarrow Y$ be any function and $g : Y \rightarrow \mathbb{R}$ bounded from above. Show that $g \circ f$ is bounded from above and that

$$\sup_{x \in X} (g \circ f) \leq \sup_{y \in Y} g. \quad (3.1)$$

In addition, we have equality when f is surjective.

Proof. The function $g \circ f$ is bounded from above whenever g is bounded from above since $(g \circ f)(X) \subseteq g(Y)$. Therefore,

$$\sup_{x \in X} (g \circ f) = \sup (g \circ f)(X)$$

exists. To prove the desired inequality, it is enough to show that $\sup_{y \in Y} g$ is an upper bound for the set $(g \circ f)(X)$. Clearly, any element of $(g \circ f)(X)$ takes the form $g(f(x))$ for some $x \in X$. Rather, it takes the form $g(y)$ for some element $y = f(x) \in Y$. Thence,

$$(g \circ f)(x) = g(f(x)) = g(y) \leq \sup_{y \in Y} g.$$

We have thus proven (3.1). Now, suppose that f is surjective. This means that $g(Y) = g(f(X))$ whence

$$\sup_{y \in Y} g = \sup g(Y) = \sup g(f(X)) = \sup (g \circ f)(X) = \sup_{x \in X} (g \circ f).$$

See the remark below for an example in which equality fails when f is not surjective. $\square$

Remark 3.1. Equality may fail in (3.1) if f is not surjective. Let $X = Y = [0, 1]$ and define functions $f : X \rightarrow Y$, $g : Y \rightarrow \mathbb{R}$ according to the formulas

$$f(x) = 0, \quad g(y) = y.$$

Then, $(g \circ f)(X) = \{0\}$ so that

$$\sup_{x \in X} (g \circ f) = 0.$$

On the other hand, $g(Y) = [0, 1]$. By an earlier exercise (Problem ??), we have $\sup g(Y) = \max g(Y) = 1 > 0$. Therefore,

$$\sup_{x \in X} (g \circ f) < \sup_{y \in Y} g. \tag{3.2}$$

4. INEQUALITIES

Proposition 6. *Let $a, b \in \mathbb{R}$ satisfy $a \leq b + \varepsilon$ for all $\varepsilon > 0$. Then, $a \leq b$.*